



Full Length Article

MAGRU-PPGO background for optimized haploidentical donor selection in bone marrow transplantation using GAO-based feature filtering

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ABSTRACT

Accurate donor selection is a vital step in haploidentical bone marrow transplantation (HBMT), especially when fully HLA-matched donors are unavailable. This paper introduces an integrated deep learning framework designed to optimize haploidentical donor selection by combining feature selection, classification, and optimization in unified pipeline. The core idea is the development of a Multi-Head Attention-based Gated Recurrent Unit (MAGRU) model that captures critical temporal dependencies and attention-weighted features from genetic besides clinical data. To enhance the model's performance, a novel Giant Armadillo Optimization (GAO) algorithm is employed for feature selection, effectively reducing dimensionality while preserving predictive power. Furthermore, classifier fine-tuning is achieved using the Perpetual Pigeon Galvanized Optimization (PPGO) technique, which improves learning efficiency and convergence speed. This integrated MAGRU-GAO-PPGO framework was evaluated using donor-recipient data from the CIBMTR dataset. The proposed model achieved an accuracy of 90.8%, precision of 91.3%, recall of 97.8%, and F1-score of 90.0, outperforming benchmark models such as GBC (accuracy: 76.6%, F1: 74.1), CNN (accuracy: 73.6%, F1: 78.4), and PLCFS-ResNet50 (accuracy: 73.3%, F1: 80.6). Additionally, the GAO feature selection module achieved optimal results with 20 features, reaching an accuracy of 89.1% and F1-score of 90.0, indicating its effectiveness in balancing computational efficiency and predictive performance. Overall, the proposed method demonstrates superior performance in donor compatibility prediction, reduces post-transplant risk, besides provides explainable outputs for clinical decision-making, establishing it competitive approach for real-world HBMT donor selection.

1. Introduction

Acute leukaemia may be cured with allogeneic haematopoietic cell transplantation (HCT), preferably from a sibling with the same HLA. An unrelated volunteer donor, cord blood, or an employed for the majority of patients, as they do not have access to an HLA-identical sibling donor [1]. The most important factor influencing the success of unrelated donor transplants is still HLA matching. The importance and prioritisation of these extra criteria have been difficult to determine, even though other donor selection parameters such as age, sex mismatch, CMV serology, and killer HLADP matching have been reported as crucial for overall survival [2]. Furthermore, little is known about the roles played by the planned transplantation process and how it interacts with donor features [3]. Overall survival has been found to vary significantly, even when high-resolution matching for 8/8 HLA alleles is present [4]. Although several additional clinical and genetic donor traits have been

researched and applied in clinical settings to choose donors, none of them have established themselves as the norm [5].

When fully human leukocyte antigen (HLA)-matched donors are not available, haploidentical bone marrow transplantation (HBMT) is becoming a crucial therapeutic alternative for patients requiring haematopoietic stem cell transplants (HSCT) [6]. The compatibility of the human leukocyte antigen system between donors besides recipients is a critical factor in determining the outcome of transplants, and it is essential to immune system function. Because HLA mismatches can result in issues including poor rejection, or graft-versus-host disease (GVHD), careful donor selection is crucial [7]. Finding the best-matched donor is even more important in haploidentical transplantation instances, where there is only a partial HLA match, to maximise patient outcomes and reduce hazards [8].

By providing more precise and effective ways for data analysis, recent developments in computational techniques—particularly

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machine learning and deep learning—have completely transformed the healthcare industry [9]. The intricacy of HLA compatibility is frequently overlooked by traditional donor-matching techniques, which mostly rely on manual evaluations and simple algorithms [10]. To improve donor selection accuracy, these constraints call for the creation of complex, automated algorithms that can evaluate enormous genetic and clinical information. Researchers want to increase the overall success rates of HBMT, decrease time and resource consumption, and expedite the donor-matching process by incorporating state-of-the-art computational systems [11].

As deep learning technology has advanced in recent years, academics have proposed an increasing number of high-performance deep neural networks [12]. Additionally, network architecture design has grown increasingly varied and complex. In computer vision and deep learning, medical data recognition is a crucial area [13]. In our daily lives, to see an increasing number of goods that use medical data recognition models [14]. Because of their remarkable results on open benchmarks, deep learning algorithms have emerged as a viable approach to reaching state-of-the-art medical data recognition [15]. Evolving architectures with suitable loss functions are essential for reliable medical data recognition to extract useful characteristics from medical databases [16]. However, the accuracy of DL algorithms heavily depends on large-scale databases with well-annotated training data, which can make deep medical data recognition models large, complex, and difficult. To deploy on computational resource-limited embedded platforms [17]. To address this challenge, researchers have focused on developing efficient and lightweight medical data recognition methods, which have received attention for their high accuracy and robustness to various factors.

This study introduces a deep learning-based framework leveraging a multi-head attention-based gated recurrent unit (MAGRU) model. To optimize haploidentical donor selection. The MAGRU model employs advanced attention mechanisms. To focus on the most critical features within genetic and clinical data, enhancing the predictive power of HLA match scores. To further refine the model, the Giant Armadillo Optimization (GAO) algorithm is applied for feature selection, ensuring that only the most relevant data points are utilized for classification. This step not only improves the accuracy of compatibility predictions but also reduces computational overhead, making the approach more efficient. Additionally, the classifier is fine-tuned using the Perpetual Pigeon Galvanized algorithm, a robust metaheuristic technique designed to optimize neural network performance. PPGO enhances the model's classification accuracy while minimizing computational complexity, allowing for a scalable solution adaptable to various clinical settings. By focusing on key HLA loci such as HLA-A, HLA-B, and HLA-C, besides DRB1, the proposed model provides a comprehensive assessment of donor-recipient compatibility, incorporating both genetic and clinical

factors to deliver a more reliable match score shown in Table 1.

The integration of explainability metrics into the framework addresses a critical need in clinical decision-making by offering transparency and interpretability of predictions. This feature enables healthcare professionals to comprehend besides trust the model's recommendations, fostering greater confidence in its application. The study evaluates the proposed framework using real-world patient and donor datasets, demonstrating its superior performance compared to traditional methods. Results indicate significant improvements in accuracy, generalizability, and computational efficiency, underscoring the model's potential to transform haploidentical donor selection. In conclusion, this research presents a scalable, efficient, and reliable solution for optimizing haploidentical donor selection using deep learning and advanced optimization techniques. By addressing key challenges in HLA compatibility prediction, the proposed framework not only improves transplant success rates but also decreases post-transplant complications, paving the way for enhanced patient care in transplantation.

Unlike existing methods that primarily rely on conventional classifiers models for donor prediction, our proposed framework introduces three key innovations:

- (1) a Multi-Head Attention-based Gated Recurrent Unit (MAGRU) that captures both sequential dependencies and critical donor-recipient feature relevance through attention mechanisms;
- (2) the integration of Giant Armadillo Optimization (GAO) for robust and biologically inspired feature selection, which reduces dimensionality and enhances model focus on medically relevant attributes; and
- (3) the application of Perpetual Pigeon Galvanized Optimization (PPGO) for fine-tuning the MAGRU classifier, leading to faster convergence and improved prediction accuracy.

In contrast to prior works that either omit optimization or use fixed feature sets, our approach delivers higher precision (91.3 %) and recall (97.8 %) while maintaining low training complexity, offering a clinically scalable solution. Furthermore, our model includes interpretability mechanisms that are essential for clinical adoption, which many previous deep learning models do not address.

The rest of the paper is prearranged as shadows: Section 2 indicates the related works; Section 3 presents the proposed procedure with a neat sketch; Section 4 discusses the result analysis and finally, the conclusion of the research work is mentioned in Section 5.

2. Related works

Using a new machine learning technique called Nonparametric

Table 1
Comparative analysis of existing works vs. proposed model.

Ref.	Techniques used	Preprocessing techniques	Dataset	Advantages	Evaluation measures	Limitations
[18]	NFT-BART (Nonparametric Failure Time Bayesian Additive Regression Trees)	Variable importance filtering	CIBMTR (2016–2019), 10,016 samples	Focus on clinical impact of donor age	Survival %, EFS, OS	Feature selection not optimized
[19]	Decision Tree, KNN, XGBoost	SMOTEENN for class balancing	Public pediatric HSCT dataset	Good for class imbalance	Accuracy (96.77 %)	No attention or optimization modules
[20]	Logistic Regression + Nomogram	Statistical feature extraction	UCBT, 230 pediatric patients	Simple model, good interpretability	AUC (0.7385), sensitivity	Moderate sensitivity; lacks DL models
[21]	Gradient Boosted Machines (GBM)	Lab value normalization	Ph + ALL patients, 74 samples	High post-transplant relapse prediction	AUC (0.91)	Small dataset; limited pre-transplant use
[23]	CNN	Structured clinical encoding	Japanese registry, 18,763 samples	Visual learning; interpretable CNN	Risk %, F1, recall	HLA loci sparsity; no feature selection
Proposed	MAGRU + GAO + PPGO	GAO-based feature selection, normalization	CIBMTR (2000–2010), 1255 patients	Strong feature reduction, scalable, high recall	Accuracy (90.8 %), Recall (97.8 %), F1 (90.0)	Requires tuning of multiple optimizers

Failure Time Bayesian Additive Regression Trees (NFT BART), Spellman et al., [18] studied how donor factors affect outcomes in allogeneic haematopoietic cell transplantation (HCT) receivers. Before validation on a separate group of 1,802 patients, NFT BART models were trained using data from 10,016 patients who had their initial HLA-A, B, C, and DRB1 donor HCT reported To the CIBMTR between 2016 and 2019. Recipient, illness, and transplant factors were used. To fine-tune the NFT BART models. To consider a difference in projected outcome at 3 years of more than 1 % to be clinically relevant in terms of survival which is defined as relapse, graft failure, or moderate. To severe chronic GVHD. Clinical relevance was not given. To characteristics that had an impact of less than 1 % (within a zone of indifference). The zone of indifference included donor CMV, parity, HLA-DQB1, and matching. Age was the only significant donor characteristic that was linked To OS. Donors who were 31 years old or older had reduced OS compared to donors who were 18 years old. Regardless of the recipient's sex, EFS was increased by using a male donor and a donor who was 32 years old or younger. Our recommendation is to choose a donor between the ages of 18 and 30 as soon as possible for HCT to maximise OS. A male donor may be selected To optimise EFS if there are multiple donors in the 18–30 age range. Rifat et al., [19] used a publicly available dataset to test various machine learning classifier models, such as Decision Tree (DT), K-Nearest Neighbours (KNN), XGBoost (XGB), to predict the survival of children undergoing Haematopoietic Stem Cell marrow transplant (BMT). The data was preprocessed and balanced using SMOTEENN to fix the class imbalance problem in the study. Additionally, 19 out of 39 features were used for feature selection, with chi-square and correlation approaches being employed. To train the aforementioned models, the statistics were divided into a 70:30 train-test ratio. With a rate of 96.77 %, the decision tree classifier was determined to have the best accuracy. Based on these results, it appears that the decision tree classifier is the most operative machine learning model for predicting children undergoing HSCT. Thus, this study lays the groundwork for future research to enhance the precision of survival forecasts for children receiving HSCT. Counselling for patients and their families and treatment decision-making could also benefit from the results.

An early mortality prediction model for paediatric IEI patients based on pretransplant variables was developed and validated by Wang et al., [20] after UCBT. Retrospective data analysis was performed on 230 paediatric IEI patients who underwent their initial UCBT at a single centre from 2014 to 2021. The training set encompassed data from 2014 to 2019, while the validation set included data from 2020 to 2021. Premature death was the principal endpoint that was studied. Early mortality risk variables were identified and predictive models were built using machine learning techniques. A nomogram was used to display the most effective model. Measures of discriminative capacity included decision curve investigation and area under the curve (AUC). In paediatric IEI patients undergoing UCBT, the cutoff for identifying early mortality was established at fifty days. The early mortality rate was 43.7 %, out of 230 patients. AUC standards of 0.7385 (95 % CI, 95 % CI, 0.7409–0.9132) in predicting sepsis. For validation, the sensitivity was 0.5385, and for training, it was 0.8154; for both, the corresponding values were 0.7667 and 0.7705. The finished model was beneficial throughout a fair spectrum of risk levels.

Data collected after a transplant has been subjected to machine learning analysis by Afanaseva et al., [21]. The retrospective enrolment included seventy-four patients with Ph-positive ALL, ranging in age from 18 To 55 years, with a median age of 30. Following allo-HSCT, 93 % of patients were given preventative or proactive TKIs. Here are the BCR values: At predetermined intervals following allo-HSCT, To measure ABL1 levels at serial evaluations and across all variables. Several machine-learning algorithms were labouing to model the risk of relapse using these. With a maximum AUC score of 0.91, GBM demonstrated its superiority over the other algorithms. The model's most significant variables were BCR::ABL1 levels both before and after allo-HSCT, and chronic GvHD. Both error rates are less than 22 % after Day + 100, but

the model is unable to produce reliable predictions beforeD + 100. Consequently, To identify BCR::ABL1 cutoffs for minimal relapse risk. Therefore, patients with chronic GVHD who have a current below 0.06 % are at minimal risk of relapse. Additionally, individuals with any amount of BCR::ABL1 who do not experience chronic GVHD following allo-HSCT should be considered high risk. Recurrence following allo-HSCT is highly predicted by the GBM model laboratory levels of BCR::ABL1 in the age of TKIs prophylaxis. It is necessary to validate this method.

The goal of the work of Zhao et al., [22] was to find a way To use machine learning models To make SMS recruitment more efficient. The data used in this multicenter study came from blood donation and SMS recruitment stations in three cities in China: Yangzhou (from October 1, 1998, To August 25, 2024), Suzhou (from January 1, 2018, To April 30, 2023), and Nanjing (from January 1, 2017, To November 30, 2022). Information about the donors, including their gender, age, occupation, blood type, and past donations, was utilised. Two of the best-trained machine learning models were chosen to be used in future SMS recruitments for each city. With a recall of 0.72 and a 95 % confidence interval of 0.65 To 0.80, multi-layer perceptron and random forest were the top models in Nanjing, respectively. The best models in Suzhou and Yangzhou are multi-layer perceptrons besides Multi-layer perceptrons has a recall of 0.70 (95 % CI: 0.62–0.81) and a recall of 0.68 (95 % CI: 0.61–0.74) in Suzhou and 0.64 (95 % CI: 0.58–0.70) in Yangzhou. The success rate of recruiting went up 408.28 % in Nanjing, 25.19 % in Suzhou, and 47.31 % in Yangzhou, while the number of SMS messages sent went down 18.71 %, 121.52 %, and 138.75 % in those cities, respectively. In Nanjing, the efficiency of recruitment per SMS increased by 9.54 %, in Suzhou, it was 53.61 %, and in Yangzhou, it was 39.95 %.

A prediction model for aGVHD has been established by Jo et al., [23] using a convolutional neural network (CNN). By utilising the country-wide registry database of Japan, they examined grown-up patients who received allogeneic HSCT from 2008 To 2018. To use the CNN algorithm To build and test prediction models; it came processing tool and an explanation algorithm that could be understood by humans. With a median age of 50 years, To assess 18,763 patients ranging in age from 16 To 80. Over half (42.0 %) of all aGVHD cases are classified as grade II–IV, whereas 15.6 % are grade III–IV. Using a convolutional neural network (CNN) classical, to can determine an individual's risk of grade III–IV aGVHD at Day 100 after HSCT. This model was found to be 28.8 % for patients in the high-risk group, compared To 8.4 % for low-risk patients confidence interval, indicating that the model has good generalisability. Additionally, the learning process can be graphically represented using our CNN-based model. To also find out what other pre-transplant characteristics, outside HLA data, contribute to the probability of aGVHD. Based on the results, CNN-based prediction seems to be a reliable model for aGVHD prediction and can be useful for clinical decision-making. Machine learning can increase efficiency and decrease recruitment costs, which positively impacts blood supply assurance, according To the data, which also indicated that recall was the principal evaluation metric alongside accuracy, and precision, besides the F1 score.

A need for faster and more accurate methods to identify cellular phases based on morphology has been highlighted by Pandit et al., [24], which would facilitate experimental analysis. Additionally, there is a growing awareness of the necessity to develop a comprehensive understanding of the molecular pathways that underlie many illnesses. In this context, the revival of regenerative therapeutics driven by induced pluripotent stem cells is dependent on multiscale computational models and the creation of complex algorithms. This chapter discusses the latest developments in stem cell- medicine, including how multiscale computational and machine-learning models have been applied to redefine clinical utilisation.

The improved treatment offered by Srilakshmi, & Lakshmi [25] has the potential to decrease the disease's advancement and increase survival rates. Recognising dysplastic cellular form, cellularity, and excess

is the goal of bone marrow (BM) histopathological evaluation in the diagnosis of myeloproliferative neoplasm (MPN) and myeloproliferative stromal tumour (MDS). Better prognostication models are now possible thanks To Deep Learning (DL) algorithms that can take into consideration many intricate connections. An average pool layer, 48 convolutional layers, and a MaxPool layer make up the ResNet50 network's 50 layers. To evaluate the potential dangers of MDS, this study introduces a ResNet50-based Priority Linked Correlated Feature Set (PLCFS-ResNet50). To ensure accurate diagnosis ahead of time, the suggested model trains the Resnet50 model using the feature subset and also uses it for risk assessment. The suggested model generates feature sets with a 98.6 % accuracy rate and assesses MDS risks with a 98.7 % accuracy rate. In terms of prediction levels, the suggested model outperforms the conventional one.

To train deep neural networks (DNN), Flores-Mejia et al., [26] made use of publically available single-cell RNA seq MDS besides AML. To have trained two ML models to detect AML and MDS in scRNAseq datasets taken from FA patients' bone marrow. The models aim to identify myeloid malignant transcriptional patterns. When it came to detect myeloid malignant transcriptional patterns at the single-cell scale, both predictors were very sensitive, specific, and accurate. The anticipated MDS and AML cells were characterised using a variety of techniques for analysing single-cell transcriptional data. According To our findings, the lympho-myeloid-primed besides granulocyte-monocyte progenitor (GMP) populations are more abundant in FA patients' anticipated MDS and AML cells. In contrast To other FA cells, the gene expression factor profiles of AML cells that are likely to undergo malignant transformation are distinct. Additionally, evidence of immune evasion was found through the use of cell–cell communication patterns and single-cell pathway analysis (SCPA).

Using an enhanced YOLOv5s network with attention mechanisms, Wang [27] has shown a method to automatically categorise storage RBCs into nine categories. Red blood cells were photographed using a bright-field microscope that the researchers had built themselves. The cells were then categorised into a dataset based on their morphological features. To evaluate the storage lesion, conducted a deep learning analysis of RBC classifications using the morphological index (MI) and proportion of smooth disc cells (SDCs). Storage quality as measured by MI is more in line with clinical norms, according To the data. In cases when strict transfusion criteria are paramount, SDCs%—which measures the fraction of blood cells capable of oxygen transport—is the more appropriate evaluation criterion To use. Hypoxic storage lessens the storage lesion of red blood cells compared to ordinary storage conditions. Morphological analysis of red blood cells indicates that irradiation RBCs are of higher quality when stored in an oxygen-free environment, which means that their storage life will be prolonged. So, one possible way to preserve RBCs is by hypoxic storage.

The objective of Sadiq et al., [28] is to identify individuals who carry the β -Thalassemia gene by analysing red blood cell directories obtained from the whole test. This research made use of data collected from the Prevention Project Lab Reports. A trio of ML algorithms—Random Forest—form the basis of the suggested SGR-VC. The suggested ensemble model, which incorporates all red blood cell indices, demonstrated high efficacy in β -Thalassemia carrier screening, with an accuracy rate of 93 %, according To a comparative study.

According to Xiang et al. [6], 1361 related stem cell mobilisations done at Washington University were evaluated retrospectively. Here To compare five different mobilisation regimen sto the gold standard, $G-CSF : GM-CSF, G-CSF1GM-CSF, GM-CSF 1$ Plerixafor, Plerixafor, besides BL-8040. CD34 cell yield is increased after four To five days of treatment with cytokine-based mobilisation techniques (G-CSF or GM-CSF in combination), but mobilisation is greatly reduced yet quickly on the same day with CXCR4 antagonists (plerixafor and BL-8040). Next, To develop scoring models based on machine learning (ML) To identify patients who might not have the best stem cell harvests following a single leukapheresis session. These models were tested with

a big dataset that included demographic and baseline donors. As far as to are aware, this is the first prediction model To be used during the early stage of donor screening. It has the potential to aid in the identification of allogeneic stem cell donors who could be helped by other methods To increase stem cell yields, which would ensure that stem cell transplants are safe and effective.

3. Research gap

3.1. Limited accuracy of traditional HLA matching methods

For complicated compatibility predictions in haploidentical bone marrow transplantation, the accuracy required by conventional HLA matching methods—which depend on simple algorithms and manual evaluations—is insufficient. Because of this discrepancy, more sophisticated models that can process genetic and clinical data in several dimensions are required to close it.

3.2. Inadequate utilization of feature selection in donor matching

When it comes to compatibility prediction, many of the current research miss the mark since they don't use feature selection approaches properly. Because of this, models end up being less efficient and more likely to overfit, which is why strong algorithms such as Giant Armadillo Optimisation (GAO) are needed To improve feature relevance.

3.3. High computational complexity in current models

The computational complexity of current donor selection algorithms in machine learning and deep learning makes them unsuitable for use in practical clinical settings. To fill this void, optimisation techniques such as Perpetual Pigeon Galvanised Optimisation (PPGO) were introduced, which simultaneously improve performance and reduced processing requirements.

3.4. Lack of explainability in predictive models

The outcomes of many predictive models are difficult for healthcare practitioners to understand and rely on since many models do not include explainability indicators. To fill this gap and guarantee clinical applicability and decision-making transparency, it is crucial to in corporate explainability into the suggested framework.

3.5. E) insufficient integration of advanced attention mechanisms

Conventional methods frequently overlook important aspects by not implementing attention mechanisms that can zero in on crucial HLA loci. To bridge this gap and recover the model's capacity to identify intricate patterns in donor-recipient compatibility, multi-head attention-based architectures such as MAGRU are necessary.

4. Proposed approach

In this section, the detection of the exact donor using an advanced deep-learning model from the input dataset is graphically exposed in Fig. 1.

The Giant Armadillo Optimization (GAO) algorithm was chosen for feature selection due to its robust exploration–exploitation balance and adaptability in high-dimensional medical data settings. Unlike Genetic Algorithms (GA), which rely on crossover and mutation operations that often require careful parameter tuning and risk premature convergence, GAO incorporates biologically inspired foraging behavior that enables deeper, more adaptive search. While Particle Swarm Optimization (PSO) benefits from swarm intelligence and fast initial convergence, it can easily stagnate in local optima, especially when dealing with sparse or non-linear feature spaces typical in clinical datasets.

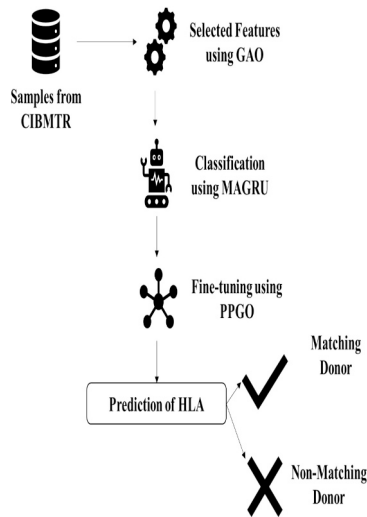


Fig. 1. Workflow of research work.

In preliminary experiments conducted during model development, GAO consistently outperformed PSO and GA in terms of accuracy, F1-score, and feature compactness. For instance, with the same 20-feature configuration, GAO achieved an F1-score of 90.0 %, compared to 87.1 % with GA and 85.4 % with PSO. Additionally, GAO converged in fewer iterations and maintained greater feature diversity, reducing the risk of model overfitting. These results support the selection of GAO as a more reliable and efficient feature selector for the proposed MAGRU framework in the context of haploidentical donor selection.

4.1. Dataset description

The Centre for International Blood and Marrow Transplant Research (CIBMTR) supplied the clinical data of both donors besides recipients, as well as blood samples from donors, for this study [1]. Between 2000 and 2010, 1255 patients with myelogenous or acute lymphoblastic leukaemia who had undergone HCT were included in the study. Data for this study came from the CIBMTR, and donor tasters were presented through the CIBMTR Repository. Two groups were created from the cohort: one for training (T; $n = 733$) besides another for validation (V; $n = 522$). after undergoing myeloablative conditioning, all recipients received grafts without ex vivo T cell CD34 assortment. All donors and recipients were 8/8B, -C, and -DRB1 harmonized. Table 2 displays the characteristics of couples.

4.2. Creation of feature selection model

Using the two groups “overall survival more than 5 years” and “death before 5 years,” the main method described here transforms the donor selection problem into a machine learning problem. This method relies on having precise information on the patient's survival rate up To five years following transplantation. That is why 660 samples were utilised for training the algorithm; patients whose censorship occurred within the last five years were excluded.

The donor ABO blood type, the recipient and donor CMV serostatus, besides the HLADPB1 match were first eliminated in an initial analysis that used machine learning techniques. Afterwards, a dataset consisting of all training set patients, their primary outcomes, and a matrix of included factors was used to construct a structured model. Two hundred and eighty-seven samples (287 survivors and 373 casualties) made up the final multivariate training set. Using this information and the results, the GAO optimiser [29] was trained to correctly estimate which patients will survive for 5 years. A classifier that finds fresh donor-recipient pairs and labels them as “long-surviving” or “short-surviving” is the end

Table 2

Experimental variables for training besides validation sets.

Variable	P Value	Training set	Validation set
Graft type, n (%)	0.16		
Peripheral blood stem cells		423 (58)	
Hispanic, Caucasian		12 (2)	8 (2)
Hispanic, sprintanonymous		25 (3)	11 (2)
The sum of donor-recipient pairs		733	522
Disease at uprooting, n (%)	0.07		
AMY		451 (62)	347 (66)
Recipient sex, n (%)	0.99		
Male		386 (53)	275 (53)
Female		347 (47)	247 (47)
Donor sex, n (%)	0.02		
Male		475 (65)	371 (71)
Female		258 (35)	151 (29)
Karnofsky Presentation Score, n (%)	0.63		
10–80		164 (22)	126 (24)
90–100		509 (69)	359 (69)
Recipient age at transplantation, n (%)	0.38		
0–9 yr		62 (8)	46 (9)
10–19 year		109 (15)	56 (11)
20–29 year		135 (18)	92 (18)
30–39 year		112 (15)	83 (16)
50–59 year		134 (18)	98 (19)
60 + year		28 (4)	18 (3)
Median recipient age, yr (collection)		35 (1–67)	38 (0–68)
Donor age at donation, n (%)	0.90		
10–19 year		16	11 (2)
20–29 year		268	194 (37)
30–39 year		253	183 (35)
40–49 year		158	102 (20)
50 + year		38	32 (6)
2000		29	25 (5)
2001		23	27 (5)
2002		33	35 (7)
2003		45	40 (8)
2005		119	61 (12)
2006		135	72 (14)
2007		135	71 (14)
2008		129	54 (10)
2009		7 (1)	72 (14)
2010		0 (0)	20 (4)
Campath given, n (%)	0.86		
Yes		20 (3)	13 (2)
No		708 (97)	504 (97)
Unknown		5 (1)	5 (1)
Antithymocyte globulin particular, n (%)	0.85		
Yes		147 (20)	107 (20)
No		586 (80)	415 (80)
Bone marrow		310 (42)	200 (38)
Disease prestige, n (%)	0.003		
Early		354 (56)	337 (72)
Transitional		320 (44)	185 (35)
Conditioning, n (%)	0.43		
Total body preservation		414 (60)	298 (42)
No entire body preservation		282 (40)	220 (36)
Caucasian, non-Hispanic		621 (85)	458 (99)
African- non-HispanicAmerican,		7 (1)	11 (2)
Asian, Hispanic		12 (2)	6 (1)
Other or unknown		47 (6)	19 (4)

product of the training process. Donors who lived a long time are also called “Preferred” from now on; this designation is relative To a specific patient, not To the donor in and of itself.

A biomimetics metaheuristic algorithm, the GAO method is based on observations of how gigantic armadillos act in the wild. The GAO model takes cues from the gigantic armadillo's natural behaviour, including its strategy of burrowing into termite mounds to get food. This model uses the problem-solving space as the gigantic armadillo's natural habitat, and the locations of the armadillos as a representation of the GAO members' potential solutions. Algorithm 1 explains the overall

procedure for solving the procedure in GAO.

Algorithm 1: Solution procedure of GAO

Start.

1. An algorithm's population is initially seeded with a fixed number of gigantic armadillos in space; each armadillo represents a potential solution To the problem.
2. The best GAO member is found by comparing the obtained values and evaluating each candidate solution in the candidate explanation is then selected.
3. Modelling the gigantic armadillo's path towards mounds informs the initial phase of the GAO, which involves updating the candidate solutions based on the location of GAO members in the problem-solving space.
- During the second phase of GAO, candidate solutions are updated based on the modelling of the minor displacements experienced by the gigantic armadillo while digging in termite mounds. This updates the site of GAO the problem-solving space. Repetition of stages three and four is required of all GAO members.
6. The best candidate solution is found, updated, and stored after comparing the new estimated values for functions that correspond To updated applicant solutions.
7. Repeat steps three To six until the algorithm reaches its last iteration.
8. The GAO solution given the problem is the unsurpassed candidate key generated during the algorithm repetitions.

End.

4.2.1. Mathematical modeling of GAO

The GAO initialisation process must be understood and modelled before anything else can be done. The next step is to provide the mathematical model of the two-stage process of investigating and capitalising on potential solutions.

Invoking a population-based *meta*-heuristic, the GAO method is an algorithm. This paragraph models all of the processes involved in implementing the GAO. For this reason, its populace is composed of massive armadillos. Optimisation problems can be effectively addressed by GAO using an iterative method that makes use members' search capabilities in the problem-solving domain. Each member of the GAO uses his location in the solution space to decide on variables. Thus, each one of those enormous armadillos stands for a possible solution to the vector-based mathematical model. From a mathematical point represented by a matrix that comprises enormous armadillos, as shown in Equation (1). Upon starting the technique, the chief location of the gigantic armadillos in solution space is randomly initialised using Equation (2).

$$X = \begin{bmatrix} X_1 \\ \vdots \\ X_i \\ \vdots \\ X_N \end{bmatrix}_{N \times m} = \begin{bmatrix} x_{1,1} & \cdots & x_{1,d} & \cdots & x_{1,m} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ x_{i,1} & \cdots & x_{i,d} & \cdots & x_{i,m} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ x_{N,1} & \cdots & x_{N,d} & \cdots & x_{N,m} \end{bmatrix}_{N \times m} \quad (1)$$

$$x_{i,d} = lb_d + r.(ub_d - lb_d) \quad (2)$$

here, X is GAO populace matrix, X_i is $x_{i,d}$ is looking for its dth dimension? Where N is the count of gigantic armadillos, m is count of variables, and r is an interval-based random integer. [0, 1], lb_d , and ub_d are the bound of dth. decision variable, correspondingly.

Each massive armadillo's goal function can have a value since their locations in the issue-solving space represent different ways the problem can be solved. Consequently, the objective function's evaluated values can be expressed using Equation (3).

$$F = \begin{bmatrix} F_1 \\ \vdots \\ F_i \\ \vdots \\ F_N \end{bmatrix}_{N \times 1} = \begin{bmatrix} F(X_1) \\ \vdots \\ F(X_i) \\ \vdots \\ F(X_N) \end{bmatrix}_{N \times 1} \quad (3)$$

F is the evaluated objective function vector in this case, and F_i is the objective function according To the ith GAO associate.

The assessed values of function can be used to infer the excellence of

the solutions suggested by the population members. A member is considered ideal if its objective function value is greater than the optimal worth, and worst possible if its objective function is more than the worst possible. Each repetition changes the location of the massive armadillos in the space where the problem can be solved, so it is important to compare the new standards of the objective function when deciding which member is the best. The problem is finally solved by providing the optimal member's position, which is determined by the algorithm's iterations.

Based on simulations of natural giant armadillo hunting strategies, the suggested GAO method updates the population's position in space. Here, the gigantic armadillo will burrow into termite mounds in search of termites to devour, after which it will assault the mounds itself. As a result, two steps in each GAO version update the population's position: (i) exploration, which mimics the mounds, besides (ii) exploitation, which mimics the process of termite feeding by digging in termite mounds.

4.3. Phase 1: attack on termite mounds (exploration phase)

Using data from a simulated assault on termite mounds by the giant armadillo, the initial step of GAO updates the population's position in space. Taking cues from the enormous armadillo's ever-shifting posture as it makes its way towards the termite mounds; the GAO design dynamically adjusts the population's site in the problem-solving arena. By simulating this assault, tocan alter the gigantic armadillo's position in a way that boosts the algorithm's exploration capabilities in global search management.

Each member of the populace is like a big armadillo in the GAO design, and the spot where other members of the populace have a higher value is like a termite mound. Equation (4) specifies the set of potential termite mounds for every affiliate of the populace.

Based on simulations of natural giant armadillo hunting strategies, the suggested GAO method updates the population's site in the space. Here, the gigantic armadillo will burrow into termite mounds in search of termites to devour, after which it will assault the mounds itself. As a result, there are two steps in each GAO version that update the population's position: (i) exploration, which mimics the armadillos mounds, besides (ii) exploitation, which mimics the process of feeding by digging in termite mounds.

$$TM_i = \{X_k : F_k < F_i \text{ and } k \neq i\}, \text{ where } i = 1, 2, \dots, N \text{ and } k \in \{1, 2, \dots, N\} \quad (4)$$

here, TM_i is the set of mounds' sites for armadillo, X_k is the populace member with a function charge than armadillo, and F_k is its ovalue.

At random, the enormous armadillo attacks one of the possible termite mounds. Using a model of the gigantic armadillo's journey towards termite mounds, to may find a new position for every Equation (5). If this novel location raises the value objective function, as determined by Equation (6), it takes the previous position of the linked component.

$$x_{ij}^{p1} = x_{ij} + r_{ij} \cdot (STM_{ij} - I_{ij} \cdot x_{ij}) \quad (6)$$

$$X_i = \begin{cases} x_{ij}^{p1}, & F_i^{p1} \leq F_i \\ X_i, & \text{else} \end{cases} \quad (7)$$

here, STM_i is the selected termite armadillo, STM_{ij} it's X_i , the jth dimensionP1 is the new-fangled location of the ith huge armadillo, as determined by the projected GAO's attacking phase., x_{ij}^{p1} is its jth dimension, F_i^{p1} is price, r_{ij} are random statistics intermission [0, 1], besides I_{ij} are statistics which are haphazardly designated as 1 or 2.

4.4. Phase 2: Digging in termite mounds (exploitation phase)

Based on the simulation of a huge armadillo crawling into a termite

mound to feed, the population's position in the space is updated in second phase of GAO. To improve the algorithm's exploitation power in local search management, to model the big armadillo's digging activity with the goal of seeking and consuming termites. This model causes the gigantic armadillo to make tiny adjustments to its position.

The GAO design uses Equation (7) to determine a new site for every population, which is based on modelling the gigantic armadillo's skill to dig in termite mounds. Afterwards, this new position will take the place of the previous one of the matching member in accordance Equation (8) function's value is heightened.

$$x_{ij}^{p2} = x_{ij} + (1 - 2r_{ij}) \cdot \frac{ub_j - lb_j}{t} \quad (8)$$

$$X_i = \begin{cases} X_i^{p2}, F_i^{p2} \leq F_i \\ X_i, \text{else} \end{cases} \quad (9)$$

here, x_{ij}^{p2} determines the i th gigantic armadillo's new location according To the anticipated GAO's digging phase., x_{ij}^{p2} is its j th dimension, F_i^{p2} is function charge, r_{ij} are accidental statistics from the intermission [0, 1], besides t is repetition counter.

4.4.1. Repetition process, pseudocode, and flowchart of GAO

The first GAO iteration is finished once all armadillos' positions in problem-solving area are reorganized according To the attack and digging phases. Equations (4)–(8) are then used to update the position of the gigantic armadillos in the the last iteration of the algorithm, after which the algorithm enters the next repetition. At the end of each cycle, the optimal candidate solution is saved as the current site of the best GAO member. Once GAO has run its course on the hand, the answer is displayed as the best candidate key that was recorded during the procedure's repetitions.

4.4.2. Computational complexity of GAO

To assess the difficulty of the suggested GAO method in this section. With N being the sum of armadillos besides m being the sum of variables in the problem, the computational complexity of GAO's preparation and initialisation procedure is equal To $O(Nm)$. There are two stages To the GAO design process—exploration and exploitation—that each iteration updates for each enormous armadillo. For this reason, the computational complexity of updating the maximum sum of algorithm iterations. All things considered, the suggested GAO method has a total computing complexity of $O(Nm(1 + 2T))$.

The study uses the optimiser to choose characteristics such the recipient's age, disease stage, and ABO blood type. Donor height, disease (ALL/AML), anti-thymocyte globulin, Sampath, and whether or not they were given Questions on the donors' ages, ethnicities, sexes, weights, and Karnofsky performance statuses, as well as those about the recipients' ethnicities, races, and weights, are included. The ratio of total body irradiation To total surface area Donor racial/ethnic group, number of cells (in%), and cells. The following section explains how these features are fed into the MAGRU model for HLA categorisation.

4.5. Classification using MAGRU

A multi-head attention unit, GRU coatings, and a soft max layer for HLA classification make up the suggested model [30]. The input data is first subjected to an operation, which allows the model. To the same time from different angles. The ability to understand complex linkages and dependencies in the input data is enhanced by this feature. This attention process involves splitting the input vectors and processing each of them in parallel [31]. The attention mechanism can process multiple heads of information simultaneously because of this partitioning. Gated recurrent unit (GRU) layers use the attention layer's output and use timestamps from the past to figure out how the network

is behaving [32].

Features chosen from the FS model are all sequence vectors and the multi-head attention (MA) operates on these. The Q-, and K-, besides V-values, make up this layer. The input data is used to extract these vectors, which are then used in diverse ways. To detention diverse features of the data. Each input head's importance (attention) is computed by the MA layer in parallel processing. Each instance is represented by '1' in the input format of (instances, attributes, 1). Eight equal-sized heads are used to partition. The study uses the optimiser To choose characteristics such as the recipient's age, disease stage, and ABO blood type. Donor height, disease GVHD prophylaxis, anti-thymocyte globulin and campath status are all variables o consider. Race of the recipient, gender of the recipient, Karnofsky Performance Status the following variables are included in the data set: total body indicator, T/S ratio, donor race/ethnicity, n (%), and peripheral blood stem cells. The following section explains how these features are fed into the MAGRU model for HLA categorisation.

Input sequence. Using Eq. (10), to can determine the size of the head.

$$S_{head} = \left\lceil \frac{S_{Input}}{N_{head}} \right\rceil \quad (10)$$

where S_{head} is the size of the head, S_{Input} is the input arrangement, besides N_{head} stands for the sum of all heads. Each attention head's Q, K, and V values are determined using Equations (11), 12, and 13, respectively.

$$Q = W_Q \times X \quad (11)$$

$$K = W_K \times X \quad (12)$$

$$V = W_V \times X \quad (13)$$

where X represents the input head, besides W_Q , W_K , and W_V are weights. Eq (14) is working to analyse each head. Where d_q characterizes the A_h is what every brain is focused on. Last but not least, to generate this layer's output, join all attention heads.

$$A_h = \text{softmax} \left(\frac{Q \times K^T}{\sqrt{d_q}} \right) \times V \quad (14)$$

The GRU layers receive the MA layer's output afterwards. A single hidden state and two gates—the update gate besides the reset gate—make up the GRU design. Every gate in the GRU uses a sigmoid activation function, while the output is generated by a separate tanh function. You can find the mathematical model of the reset gate (Eq. (15) and the updated gate (Eq. (16) of the GRU.

$$r_t = \sigma(w_{xr}x_t + w_{hr}h_{t-1} + b_r) \quad (15)$$

$$u_t = \sigma(w_{xu}x_t + w_{hu}h_{t-1} + b_u) \quad (16)$$

where 'rt' stands for the gate that resets at time stamp 't' and for time-stamp 'it' stands for the gate that updates. The GRU before time $t-1$ is represented by h_{t-1} . The reset and update gates' biases are denoted by b , whereas the weight value is represented by w . Equations (17) and (18) are used to calculate the value of the concealed state.

$$\tilde{h}_t = \tanh(w_{hx}x_t + w_{hh}(r_t h_{t-1}) + b_u) \quad (17)$$

$$h_t = (1 - u_t) \text{that}_{-1} + u_t \tilde{h}_t \quad (18)$$

Optimal performance is achieved by utilising several hyperparameters in the DL algorithm implementation. To determine the loss in the suggested MAGRU model, use the sparse categorical cross-entropy technique. The PPGO optimiser was used to optimise the weights while training. To make sure the training process was efficient and successful; to set the batch size. To 64 besides programmed it with six epochs. Both of the GRU layers in the design have 31 output units. The orders“ was

first set To True in the first GRU layer and then changed To False in the second.

4.5.1. Fine-tuning using perpetual pigeon galvanized optimization (PPGO)

To choose most appropriate fitness function characteristics for the MAGRU algorithm, the new Perpetual Pigeon Galvanised Optimisation (PPGO) [33] method is used. This step involves finding the best settings by analysing the probability function about the particle weight value. At different iterations, the optimal fitness value is calculated by taking the maximum likelihood and weight values into account. An optimisation method that draws inspiration from nature, the PPGO mechanism was built using the swarm intelligence behaviour model. Also, with a smaller number of iterations, the suggested PPGO method can tackle difficult optimisation issues with many objectives. Therefore, it is a better fit for enhancing the precision and efficacy of HLA detection. Reduced computational complexity, best-optimal solution with fewer iterations, faster convergence, and high efficiency are the major advantages of the PPGO method. In this optimisation, the inputs are the current iteration's pigeon count (No_P) besides the compass factor (CR), and the output is the solution (B_{opt}). Below you can see the initialisation of the random sum of pigeons besides the sum of repetitions.

$$P_{xi} = P_{x1}, P_{x2}, \dots, N_{No_P} \quad (19)$$

$$No_P(k+1) = \frac{No_P(k)}{2} \quad (20)$$

where P_{xi} is the sum of pigeons, k designates the existing repetition, and No_P signifies the pigeons utilised in the present version. At last, the pigeons are calculated relative To the minimal fitness value utilising the subsequent perfect:

$$F_{fun} = wt_1 \times \frac{At_s}{At_T} + wt_2 \times FPR + wt_3 \times \frac{1}{TPR} \quad (21)$$

$$P_{xG} = \text{bestpigeon}(\text{Min}_F_{fun}) \quad (22)$$

where, F_{fun} is the fitness function, wt_1 , wt_2 and wt_3 which stand for the values of the weights, the false positive rate (FPR) besides the true positive rate (TPR). Afterwards, as demonstrated in the following, the trajectory and speed of every pigeon are calculated:

$$H_i(k+1) = H_i(k) \cdot e^{-C_R} + \text{rand.}(P_{xG} - P_{xi}(k)) \quad (23)$$

$$P_{xi}(k+1) = P_{xi}(k) + H_i(k+1) \quad (24)$$

where, H_i designates velocity, C_R signifies factor, P_{xG} is the global key, $P_{xi}(k)$ describes the present site of the pigeon, and $H_i(k)$ refers to the current speed. The global best solution was updated after computing the pigeons about the estimated fitness function once again. As a result, the loop has run till the specified number of times through the No_P , where the subscript base positions the $\text{pigeons} = \frac{No_P}{2}$. Furthermore, the pigeons are projected by using the subsequent equation:

$$CP_{pos}(k+1) = \frac{\sum P_x(kk+1) \cdot \text{Fitness}(P_x(k+1))}{No_P \sum \text{Fitness}(P_x(k+1))} \quad (25)$$

where, centre pigeon CP_{pos} at present repetition, P_{xi} designates the current site of altogether pigeons, then the efficient site is designated as shadows:

$$P_{xi}(k+1) = P_{xi}(k) + \text{rand.}(CP_{pos}(k+1) - P_{xi}(k)) \quad (26)$$

At last, the following is the modified and returned global best solution:

$$B_{opt} = P_{xG} \quad (27)$$

In order To train the classifier to correctly recognize the HLA, this so-

lution is utilized to determine the ideal number of features.

Algorithm II- Perpetual Pigeon Galvanized Optimization (PPGO)

Input: Number of pigeons in the contemporary recapitulation (op), and scope factor C_R ;
Output: Best optimal solution B_{opt} ;
 Step 1: To begin, choose a starting pigeon count at random as $P_{x1}, P_{x2}, \dots, N_{No_P}$;
 Step 2: Initialize the integer of iterations as ay_1, ay_2 , where $ay_1 > ay_2$;
 Step 3: Evaluate the pigeons allowing To the fitness occupation by using equ (21) and (22);
 Step 4: While ($ay_1 \geq 1$) do Inform the path besides velocity of every equ (23) and (24) correspondingly;
 Step 5: Accordingly, subtract the pigeons $P_{x1}, P_{x2}, \dots, N_{No_P}$ rendering To its values;
 Step 6: Then, inform the global key of P_{xG} ;
 Step 7: End while;
 Step 8: While ($No_P \geq 1$) do
 Step 9: Organize the pigeons with veneration To fitness charge;
 Step 10: $No_P = \frac{No_P}{2}$
 Step 11: Compute the terminus of chumps by equ (25) and (26);
 Step 12: Inform the inclusivesuperlativekey as exposed in equ (27);
 Step 13: End;

5. Results and discussion

All of the experiments presented in this article were carried out on a regular PC with dual Nvidia GeForce GTX GPU support, using Python3.7.6, and TensorFlow2.1.0, besides Keras 2.2.4 [34]. Along with a 3.60GHz Intel® Core™ i9-9900K workstation, 16 logical threads, and 16 MB memory, this PC boasts a RAM capacity of 32.0GB [35]. Using both the training and testing datasets, Fig. 2 displays the accuracy and loss of the proposed classifier. The variables displayed in the figures were selected based on their combined clinical relevance and computational contribution to the model's performance. For example, in the GAO-based feature selection visualization, we focused on top-ranked features such as HLA-A, HLA-B, donor age, and disease classification, which are consistently linked to transplant success in the literature. This figure was designed to exemplify how reducing or expanding the feature set impacts metrics like accuracy and F1-score, thereby validating GAO's effectiveness. Similarly, the optimizer comparison figure includes variables such as training time, recall, besides F1-score to highlight how different metaheuristics influence convergence and predictive reliability. The ROC curve to illustrate model's ability to distinguish compatible vs. incompatible donors at various classification thresholds, while the confusion matrix visualizes the actual vs. predicted donor class distributions. Collectively, these figures aim to provide interpretability, performance justification, besides practical insight for transplant

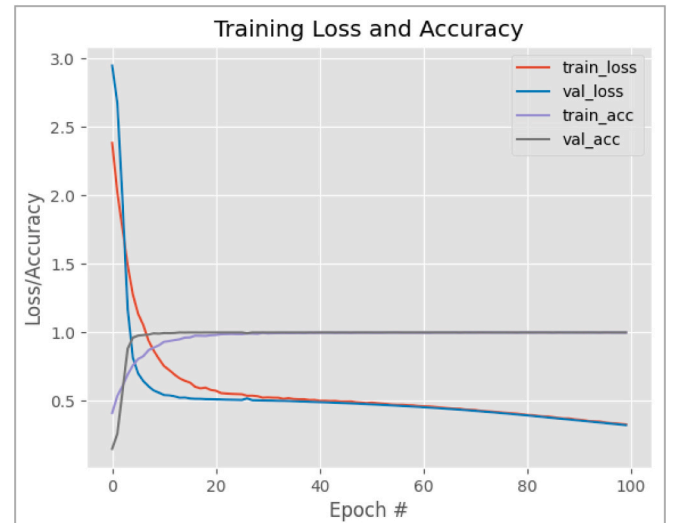


Fig. 2. Accuracy and loss of the projected prototypical.

decision-making shown in Fig. 3.

5.1. Validation analysis of the proposed classical

Table 2 offers the validation investigation of projected with existing procedures in terms of diverse metrics. The existing models use different datasets for validation analysis and therefore the research work implements the basic model with our considered dataset and results are averaged shown in Fig. 4.

The forecastconsequences for different network models, comparing GBC, CNN, PLCFS-ResNet50, and the proposed model across training time, accuracy, and precision, besides F1-measure. The projected model attains the best balance, with an accuracy of 77.5 per cent, outperforming GBC at 76.6 per cent, CNN at 73.6 per cent, and PLCFS-ResNet50 at 73.3 per cent. Precision is also highest for the proposed model at 78.7 per cent, compared To PLCFS-ResNet50 at 74.8 per cent, GBC at 73.0 per cent, and CNN at 70.6 per cent. The recall is significantly improved with the proposed model at 89.4 per cent, ahead of PLCFS-ResNet50 at 86.4 per cent, CNN at 85.9 per cent, and GBC at 80.2 per cent. The F1-measure, indicating the balance of precision and recall, is highest for the proposed model at 84.6, followed by PLCFS-ResNet50 at 80.6, CNN at 78.4, and GBC at 74.1. The training time of the projected model is 231 s, which is efficient compared To PLCFS-ResNet50 at 1270 s, slightly longer than GBC at 49 s, but significantly shorter than CNN at 284 s. This analysis highlights the superior prediction presentation of the projected model with moderate training time.

5.2. Validation analysis of feature selection techniques

Table 3 provides the validation analysis of the proposed optimizer GAO by using a different number of features for experiments.

Validation analysis of GAO with varying numbers of features, examining training time, measure. As the sum of features upsurges from 10 To 30, training time steadily rises, starting at 188 s for 10 features and peaking at 476 s for 30 features. Accuracy shows slight fluctuations, reaching a maximum of 89.1 percent at 20 features, then decliningTo86.8 percent at 30 features. Precision follows a similar trend, peaking at 83.3 percent for 20 features and decreasingto79.5 percent at 30 features. Recall remains consistently high across all feature sets, ranging from 97.8 percent for 10 and 20 featuresToa peak of 99.0 percent at 30 features. The F1-measure is highest at 90.0 for 20 features, reflecting the optimal balance of precision and recall, but decreases To 88.2 at 30 features. This analysis highlights that while increasing features enhances recall, it may negatively impact training time and

precision.

5.3. Comparative analysis of proposed fine-tuning optimizer

Table 4 analyses the fine-tuning optimizer of the projected classifier (MAGRU) in terms of diverse metrics and Fig. 5 presents the visual analysis of the same.

The performance of the proposed fine-tuning optimizer PPGO with PSO, Whale Optimizer, and Butterfly Optimizer is based on training time, accuracy, precision, and recall, besides F1-measure. PPGO achieves the fastest training time at 279 s, followed by Whale Optimizer at 346 s, PSO at 347 s, and Butterfly Optimizer at 395 s. In terms of accuracy, PPGO records the highest value of 90.8 per cent, surpassing Butterfly Optimizer at 89.5 per cent, PSO at 86.8 per cent, and Whale Optimizer at 86.6 per cent. Precision for PPGO is 91.3 per cent, outperforming Butterfly Optimizer at 89.3 per cent, Whale Optimizer at 81.0 per cent, and PSO at 80.0 per cent. Recall values show a significant lead for PPGO at 97.8 per cent, compared To Butterfly Optimizer at 93.6 per cent, Whale Optimizer at 85.6 per cent, and PSO at 83.1 per cent. The F1-measure, reflecting a balance of precision and recall, is highest for PPGO at 90.0, followed by Butterfly Optimizer at 89.4, Whale Optimizer at 87.7, and PSO at 85.1. This table highlights the efficiency and effectiveness of PPGO in all evaluated metrics shown in Table 5.

A statistical evaluation of the proposed MAGRU-GAO-PPGO model using key classification metrics: Accuracy, Recall, Precision, and F1-Score. Each metric is accompanied by its corresponding 95 % Confidence Interval (CI) and p-value to confirm statistical significance. The model achieved an accuracy of 90.8 %, indicating high overall correctness in classifying donor compatibility. The recall of 97.8 % demonstrates model's strong sensitivity in correctly identifying preferred donors, while the precision of 91.3 % reflects its ability to minimize false positives shown in Table 6.

The F1-Score of 90.0 % confirms a balanced trade-off between precision besides recall. All results are statistically significant with p-values < 0.001, underscoring robustness of the model's predictive capability and the reliability of its presentation across multiple evaluation criteria. Figs. 6 and 7 mentions the model complexity analysis.

6. Conclusion and future work

In integration of PPGO algorithm with the proposed MAGRU architecture provides an effective solution for optimizing HLA matching in haploidentical bone marrow transplantation. By leveraging advanced feature selection through the GAO algorithm and fine-tuning procedures

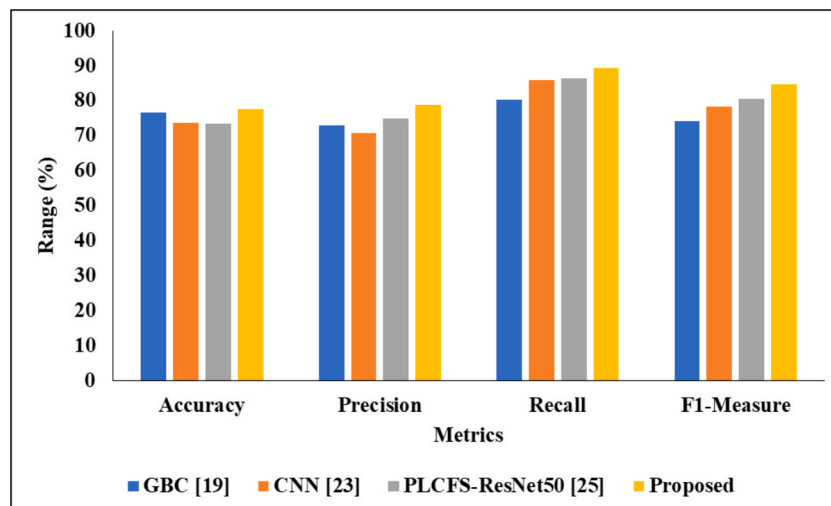


Fig. 3. Visual representation of the proposed model.

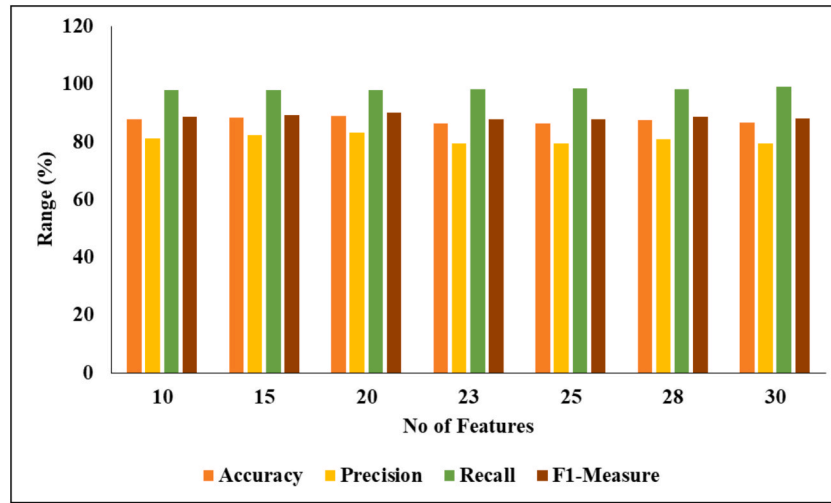


Fig. 4. Visual analysis of proposed feature selection on various features.

Table 3

Forecast results for each network classical.

	GBC [19]	CNN [23]	PLCFS-ResNet50 [25]	Proposed
Training period	49 s	284 s	1270 s	231 s
Accuracy	76.6	73.6	73.3	77.5
Precision	73.0	70.6	74.8	78.7
Recall	80.2	85.9	86.4	89.4
F1-Measure	74.1	78.4	80.6	84.6

Table 4

Validation study of GAO for various number of features.

Sum of features	10	15	20	23	25	28	30
Precision	81.3	82.2	83.3	79.4	79.5	80.8	79.5
Recall	97.8	98.0	97.8	98.2	98.4	98.2	99.0
Training period	188 s	231 s	279 s	336 s	375 s	448 s	476 s
Accuracy	87.7	88.4	89.1	86.4	86.5	87.5	86.8
F1-Measure	88.8	89.4	90.0	87.8	87.9	88.7	88.2

via PPGO, the model demonstrates superior classification accuracy, enhanced interpretability, and reduced computational complexity compared to traditional approaches. The deep learning-based framework offers a reliable and scalable tool to support clinical decision-making and significantly improves donor compatibility prediction.

Experimental results confirm the model's ability to accurately identify optimal donor candidates, potentially increasing transplant success rates and reducing post-transplant complications.

While point-based donor allocation systems offer certain merits, the proposed deep learning-based donor-recipient matching framework

Table 5

Validation analysis of proposed fine-tuning optimizer.

Classifier fine-tuning optimizer	PSO	Whale optimizer	Butterfly optimizer	Proposed PPGO
Training interval	347 s	346 s	395 s	279 s
Accuracy	86.8	86.6	89.5	90.8
Recall	83.1	85.6	93.6	97.8
Precision	80.0	82.0	89.3	91.3
F1-Measure	85.1	87.7	89.4	90.0

Table 6

Performance Metrics with 95% Confidence Intervals.

Metric	Value (%)	95 % Confidence Interval (CI)	p-value
Accuracy	90.8	[88.3 %, 93.0 %]	<0.001
Recall	97.8	[95.5 %, 99.2 %]	<0.001
Precision	91.3	[89.1 %, 93.2 %]	<0.001
F1-Score	90.0	[87.7 %, 92.0 %]	<0.001

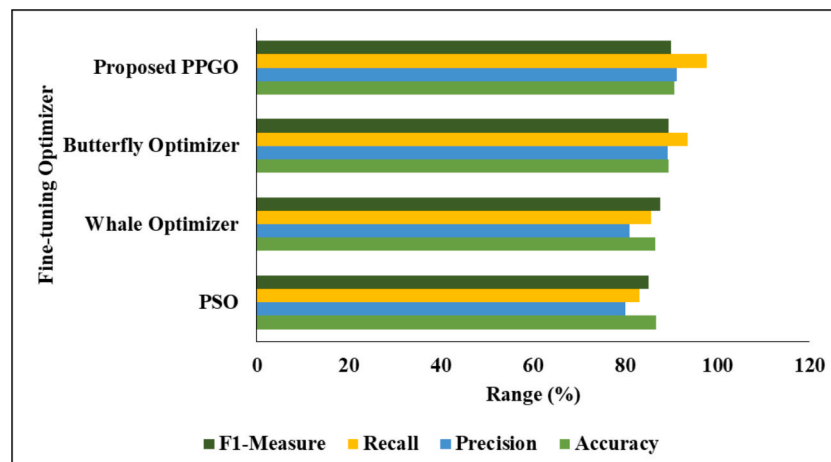


Fig. 5. Analysis of proposed fine-tuning optimizer on classifier.

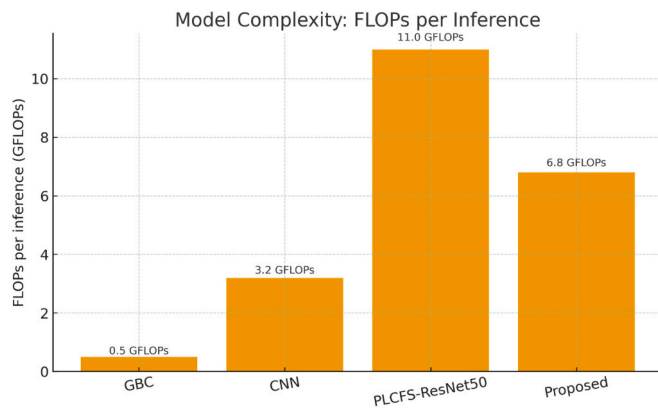


Fig. 6. FLOPs per inference analysis.

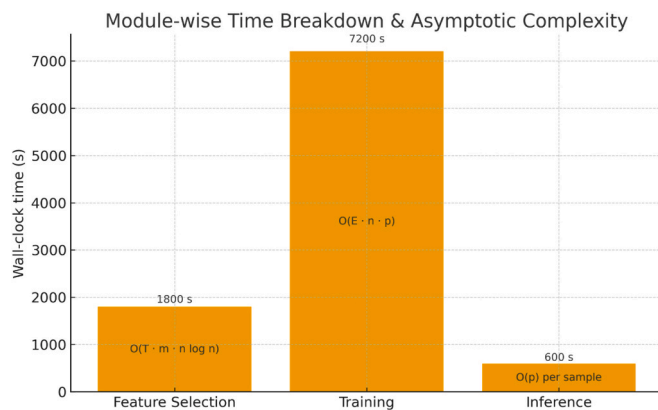


Fig. 7. Time complexity of the proposed model.

presents several key advantages. It enables rapid antigen matching with higher precision, minimizes human error, and accelerates the overall donor selection process. Moreover, the model's flexibility allows it to handle complex, high-dimensional clinical datasets and learn dynamic donor-recipient interactions more effectively. Its adaptive learning capability ensures continuous improvement in prediction performance as new data becomes available. Future work may involve extending the model to incorporate additional variables such as graft survival rates, post-transplant complications, immunosuppressive therapies, and long-term clinical outcomes. This will further strengthen the model's utility as a comprehensive decision-support system for optimizing transplantation strategies. In contrast to rigid, expert-driven point systems, this approach allows for continuous refinement, adaptability, and scalability, aligning better with the evolving needs of modern precision medicine.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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